

Diameter_analysis_Ultem(Stratasys)

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R Markdown

Filament Diameter Analysis — Ultem (Stratasys)

This document analyzes the diameter variation of a ~51m spool of Ultem filament (Stratasys) measured using a bench test rig..

Two orthogonal Hall-effect diameter sensors (TMAG5273) and a Lika A41H incremental encoder were used to measure filament diameter and length simultaneously. Sensor 1 (0x78) and Sensor 2 (0x22) are positioned 60mm and 120mm downstream of the encoder respectively.

The analysis examines diameter variation between the two sensors, ovality along the spool length, and deviation from the nominal filament diameter of 1.75mm.

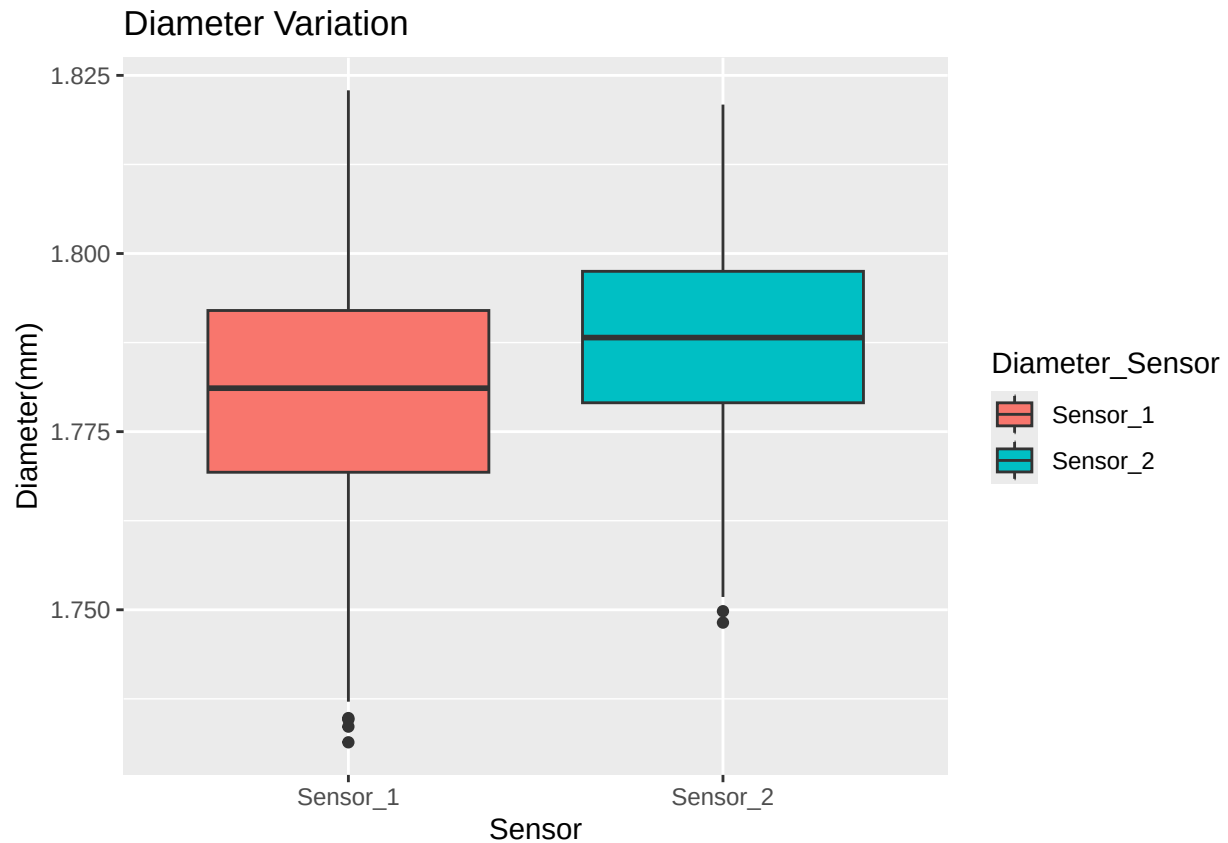
```
files<-list.files(path= "../Logged_data",
                  pattern = "*.csv",
                  full.names = TRUE)

data<- files %>%
  map(read.csv) %>%
  bind_rows()#combining all the rows

data_r<- data %>%
  distinct(length_mm,.keep_all=TRUE) %>% # removing duplicates or cases where the spool was stationary
  mutate(
    pos_sensor_1= length_mm-60, # distance between point of measurements and logged measurement for sen.
    pos_sensor_2=length_mm-120,# distance between point of measurements and logged measurement for sens
    ovality= abs(diameter1_mm-diameter2_mm),
    length_m=length_mm/1000,
    segment=cut(length_m,breaks=seq(0,52,by=5.2),include.lowest=TRUE))
```

Plot 1 — Diameter variation between sensors

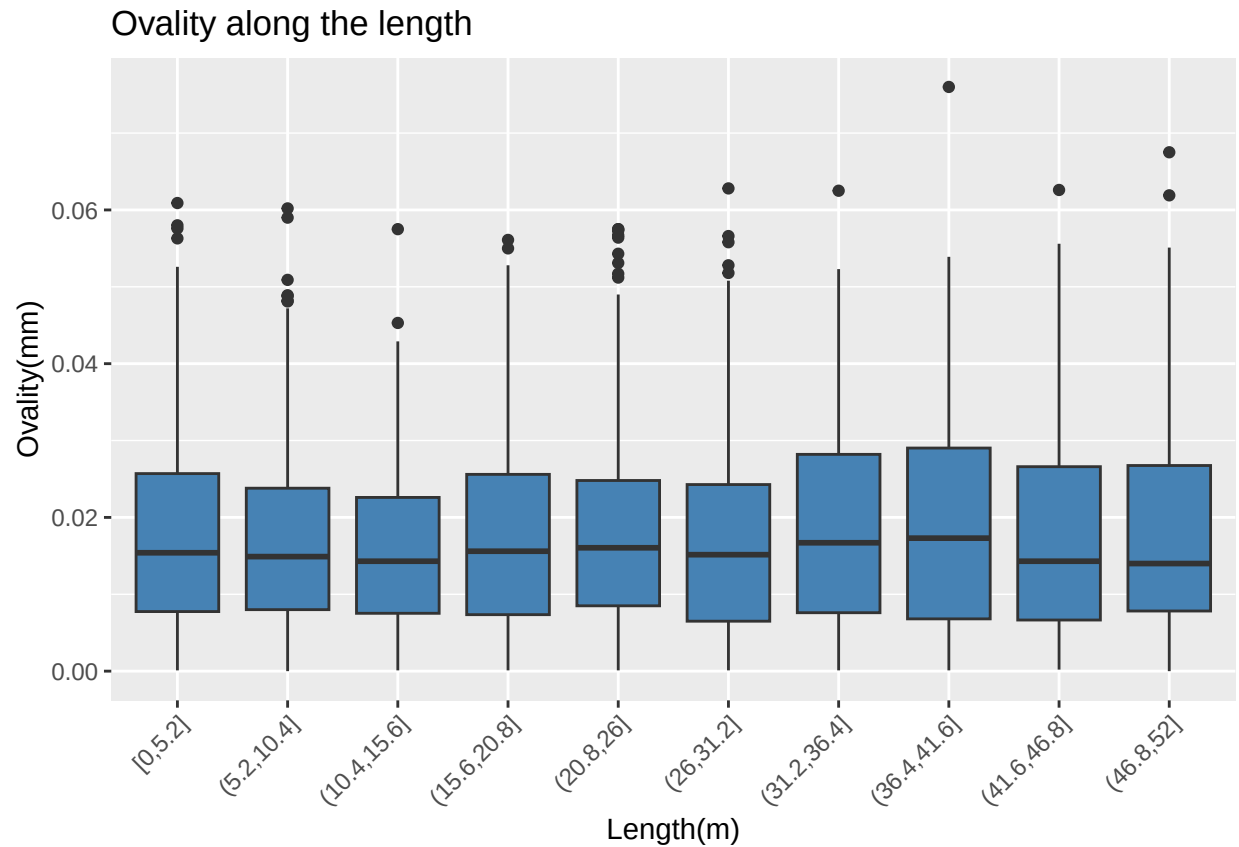
```
ggplot(data_r)+
  geom_boxplot(aes(x="Sensor_1", y=diameter1_mm,fill= "Sensor_1"))+
  geom_boxplot(aes(x="Sensor_2",y=diameter2_mm,fill = "Sensor_2"))+
  labs(title = "Diameter Variation", x="Sensor",y="Diameter(mm)",fill= "Diameter_Sensor")
```



Plot 2 — Ovality along the length of the filament measured

```
ggplot(data_r, aes(x=segment,y=ovality))+
  geom_boxplot(fill="steelblue")+
  geom_smooth()+
  labs(title = "Ovality along the length",x="Length(m)",y="Ovality(mm)")+
  theme(axis.text.x = element_text(angle=45,hjust=1))
```

'geom_smooth()' using method = 'loess' and formula = 'y ~ x'



Plot 2 — Deviation from 1.75mm

```
data_long<- data_r %>%
  mutate(
    dev_sensor1=diameter1_mm-1.75,
    dev_sensor2=diameter2_mm-1.75) %>%
  pivot_longer(cols=c(dev_sensor1,dev_sensor2),
    names_to="sensor",
    values_to = "deviation") %>%
  mutate(sensor=recode(sensor,
    "dev_sensor1"= "Sensor_1",
    "dev_sensor2"="Sensor_2"))

ggplot(data_long,aes(x=length_m,y=deviation,colour = sensor))+
  geom_point(alpha=0.2)+
  geom_smooth()+
  geom_hline(yintercept = 0,linetype="dashed",colour="red")+
  labs(title="Deviation from the diameter of 1.75mm along length of filament",x="Length(m)", y="Deviati

## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

